OF
YANKEE GRANITE

An Account of the Building of
Bunker Hill Monument

By E. H. CAMERON

BOSTON
Bunker Hill Monument Association
1953

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By E. H. CAMERON

With a Foreword by John Silsbee Lawrence

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FOREWORD

Had the British bullet that went through the hat of Major Samuel Lawrence of Groton near Bunker Hill on 17 June 1775 been aimed a bit lower, some thousand descendants of Major Lawrence would not have been born and I should not now have his dispatch wallet, nor his letter explaining the battle and mentioning General Dearborn, my first wife's ancestor. The Bunker Hill Monument might not have been built; the Massachusetts Hospital Life Insurance Company might not have been developed; various charitable institutions might not have had adequate financial support and the dams at Waltham, Lowell and Lawrence might not have been constructed to provide the power to operate textile mills. John Brown might not have been sent by the Emigrant Aid Society of Boston to Kansas, where he called his camp Lawrence, now one of the flourishing cities of Kansas. Had Major Lawrence been killed we should not have had a Lawrence as Mayor of Lowell, or as Ambassador to the Court of St. James. Two Bishops of the Episcopal Church, a President of Harvard College, the present Senator Saltonstall, and many young men and women now actively interested in the problems of our nation might never have been born and I should not have qualified for membership in the Society of the Cincinnati.

From reflecting upon the bullet that came so close to disposing of Major Samuel Lawrence at Bunker Hill I was led to the history of the monument commemorating the battle, in the building of which his son—and my great-grandfather—Amos Lawrence took a leading part. Much has been written about Bunker Hill Monument, but the splendid article by Mr. E. H. Cameron in The Technology Review *for May 1952 and June 1952 summarized its history so well that I was anxious to have it available in more permanent form. Mr. Cameron and the editors of* The Technology Review *very kindly gave permission for this reprint, and Mr. Walter Muir Whitehill, Director and Librarian of the Boston Athenæum, has seen the pamphlet through the press. Mr. Whitehill has also selected the illustrations, and has included in them a hitherto unpublished drawing, owned by the Boston Athenæum, of Bunker Hill Monument under construction in 1837.*

This was obviously the source for the somewhat inaccurate wood engraving published in The American Magazine of Useful and Entertaining Knowledge, *III* (1839), 404 *and subsequently reproduced in later accounts of the monument. The illustrations include portraits of the two largest individual contributors to the building of the monument—Amos Lawrence and Judah Touro, the Jewish merchant of Newport, Rhode Island. I am grateful to the Massachusetts Hospital Life Insurance Company and the Redwood Library and Athenæum of Newport for furnishing photographs of these portraits and permitting their reproduction, and to Mr. T. Temple Pond, President of the Bunker Hill Monument Association, for allowing this reprint to be published by the Association.*

JOHN S. LAWRENCE

OF YANKEE GRANITE

PROLOGUE

On Saturday, 17 June 1775, on a fortified hilltop farm near Bunker's Hill, Charlestown, Mass., a volunteer force of American citizens faced the professional soldiery of the world's strongest nation. When their scant supply of ammunition gave out, the survivors retired in good order, to learn later that 140 of their neighbors and other companions had been killed in the fight. Their battle is therefore registered as an American defeat. It proved to be a striking victory, however, for historians agree that the Battle of Bunker Hill set the pace that led to ultimate victory in the American War of Independence. This little force of farmers, mechanics, tradesmen, and professional men had demonstrated how Americans should fight, when their independence is threatened.

On the field where the battle was fought, the Bunker Hill Monument has now stood for over a century, the rugged lines of its granite masonry symbolizing the enduring strength of the stern spirit of American independence that it commemorates.

ABOUT 40 years after the Battle of Bunker Hill, all New England was deeply stirred by a pamphlet published by Major General Henry Dearborn who had taken part in the engagement. The pamphlet accused General Israel Putnam, one of the most revered of the Revolutionary heroes, of incapacity and cowardice in the battle. Thereupon, the Battle of Bunker Hill was fought over and over again, at the wharves, sail lofts and ropewalks of Boston, and in all places where men gathered to work and to talk about the events of the day. Crowded nine inside and five on top of the jolting four-in-hand stagecoaches from Boston, friends and foes of the popular Revolutionary hero would wrangle over his conduct at the battle. It would be a long argument, at five miles per hour, with little room for gestures. With tankards in hand, by the warm fireplace in the low-ceilinged tavern of the village where the coach would stop for the night, the passengers could express their convictions more forcefully, and the Battle of Bunker Hill would become a very live topic indeed. The furor over the Putnam-Dearborn controversy became secondary, however, as the bald fact was realized that, aside from a small wooden column, no memorial existed on the site of one of the most famous military engagements of American history.

In the good Yankee fashion a group of prominent citizens conferred over their Madeira wine and coffee on ways to correct this humiliating situation, and in the year 1823, these men formed the Bunker Hill Monument Association, to solicit private contributions sufficient to build a monument on Breed's Hill, where the battle had been fought, in the town of Charlestown, now a part of the city of Boston, Mass.[1]

[1] From the start, the site of the battle seems to have been called Bunker Hill, although it was actually fought on Breed's Hill. The probable reason for this inaccuracy is that Bunker's Hill was then 110 feet high, and the adjacent 62-foot-high Breed's Hill was considered only a spur of the higher summit. Certainly, a contemporary British military map is entitled, "A Plan of the Action at Bunker's Hill."

Unlike the Washington Monument, which had to be completed by government funds, the Bunker Hill Monument was financed practically wholly by private means. Our independent ancestors did not count much on government aid in the building of a memorial to relatives or neighbors who had died in the battle; such monuments were personal matters. Of the total collected amount of about $134,000, only $7,000—a grant from the Commonwealth of Massachusetts—came from other than private contributions. (The amount raised to build the monument is roughly equivalent to $1,000,000 in modern money.) Aside from two donations of $10,000 each, the individual gifts ranged from a few at $1,000 to many at $0.25 each. Naturally, such a scheme of financing took a long time, and 18 years elapsed before the monument was dedicated. At a critical period, the Massachusetts Charitable Mechanics Association, started by Paul Revere and others, years earlier, joined with the Bunker Hill Monument Association to raise funds and help direct operations. At a still more critical period, the women of New England held a fair which brought in over $30,000, and the completion of the monument was assured.

Surely, the record of no other national memorial provides such a true cross section of American democracy as exhibited in the roster of the rich, those of moderate means, and the poor but independent citizens whose contributions made possible construction of the monument.

The magnitude of American structures of the year 1825, when the cornerstone of the monument was laid, was largely limited by the physical strength of those who had to build them—men, horses, and oxen. To raise the huge stones of the monument to such dizzy heights was a tremendous undertaking with the crude construction methods of the day. The builders of the monument had much to inspire them to devise better methods, however, in the examples of other enterprises in this virile period of American development. Steam navigation had already made notable progress in America, and while the lower courses of the monument were being laid, the first steam locomotives began to appear on the young American railroads. Canals, waterpower developments, and many new industries were being started in the young democracy.

A great contribution of the builders of the monument to the record of achievements of this period was their demand for granite in huge quantities to build it. This demand inspired the construction of the Granite Railway at Quincy, Mass.—America's first railroad.

The story of the promotion, design, and construction of the monument is therefore doubly intriguing. It gives a vivid picture of the status of construction methods of the period, when America stood on the threshold of the age of machinery. It also reveals the spirit of audacious determination of our construction forebears as they developed their unprecedented processes from which our present marvelously efficient methods of construction have sprung. The spirit of the builders of the monument is worthy of that of the heroes of the battle, which their masterpiece has now commemorated for over a century.

The Obelisk

Few modern architects, engineers, or contractors are privileged to work in such distinguished company as did architect Solomon Willard, engineers Loammi Baldwin and Gridley Bryant, and contractor James Sullivan Savage, who designed and built the obelisk which is called the Bunker Hill

Monument. They were associated with Daniel Webster and young Edward Everett, both of whom later became Secretary of State; with Thomas Handasyd Perkins (still revered in Boston as the man who endowed the Perkins Institution for the Blind), merchant prince of Boston, with the famous artists, Washington Allston and Gilbert Stuart; and their monument followed the classic lines of the model submitted by Horatio Greenough, a Harvard student, who later became a noted American sculptor.

To the amazingly talented architects and engineers of Egypt, 50 centuries ago, the Bunker Hill Monument would have been a simple structure to design and construct. To these ancient builders, it would have appeared to be merely a somewhat stubby shaft, devoid of the beautiful, deeply carved hieroglyphic record which ornamented their own obelisks from base to pyramidion (apex), and it would be a simple thing to erect. In fact, some would say that the monument is really not an obelisk, for it is built of many stones of a few tons weight each, whereas a single stone composed a typical Egyptian obelisk. Such stones sometimes weighed as much as 500 tons. They were transported hundreds of miles and by some now unknown method were erected to the vertical position by manual labor.

Like the Egyptians, the modern engineer would also call the monument easy to design and build. Today’s light, thin-walled chimneys of comparable height, pose much harder problems of stability against wind, and their designers feel fortunate when a chimney can rest on as firm a foundation as the glacial drumlin soil of Breed’s Hill. Why, then, was the erection of the monument considered such an unusual feat at the time?

The answer is obvious: the monument was built in the days of hand labor supplemented by animal power, and hand labor to the independent Boston mechanic of over a century ago did not mean hundreds of slaves tugging in unison to the drumbeat of an Egyptian timer, while an overseer cracked his whip. And the builders had determined to construct their monument of one of the hardest of building stones—New England granite—in the use of which there was then little precedent.

When the Bunker Hill Monument Association offered a prize of $100 for the best design of a monument, many plans, mostly of columns, were submitted. The Board of Artists of the Association (Daniel Webster, Gilbert Stuart, Washington Allston, Loammi Baldwin, and George Ticknor), who had to pass upon the submitted designs, favored the Greenough model based on an Egyptian obelisk of ancient Thebes. Although the directors had strongly favored a column, they yielded to the judgment of the Board of Artists and adopted the obelisk design instead.

Upon the adoption of the successful design, a committee, of which Loammi Baldwin was chairman, was appointed to "report a design of an obelisk." Baldwin was a Harvard graduate, who had studied abroad under the patronage of Count Rumford.[2] He had become one of America's most prominent engineers. Baldwin was responsible for the construction of the dry docks at the Charlestown and Norfolk Navy Yards; planned a canal tunnel (later built as a railroad tunnel) through the Hoosac Mountain; and was active in surveys for an adequate water supply for Boston, in the day when Boston people got their water from wells. Baldwin and his associates on the committee first went to the Boston and Roxbury Milldam (now Beacon Street), from which the monument would be prominently visible across the Charles River. Miniature models of different dimensions were mounted on the Milldam fence and were viewed from a definite distance to the rear. In this highly practical manner, the size of the most striking monument on distant Bunker Hill was determined.

[2] The famous scientist, Benjamin Thompson, of Woburn, Mass., who later became an English citizen, and who established the fact that heat is a form of motion.

The Baldwin Report on the design of the Bunker Hill Monument, described as neatly handwritten, was one of the valuable documents in the literature of early American engineering history. It ranks with the "Private Canal Journal" of DeWitt Clinton,[3] who promoted the Erie Canal; the report on American railroad standards of 100 years ago by Captain (later General) George B. McClellan of Civil War fame; Roebling's report on the proposed Brooklyn Bridge; and similar historical documents that describe the methods from which the present processes of promotion and construction have sprung.

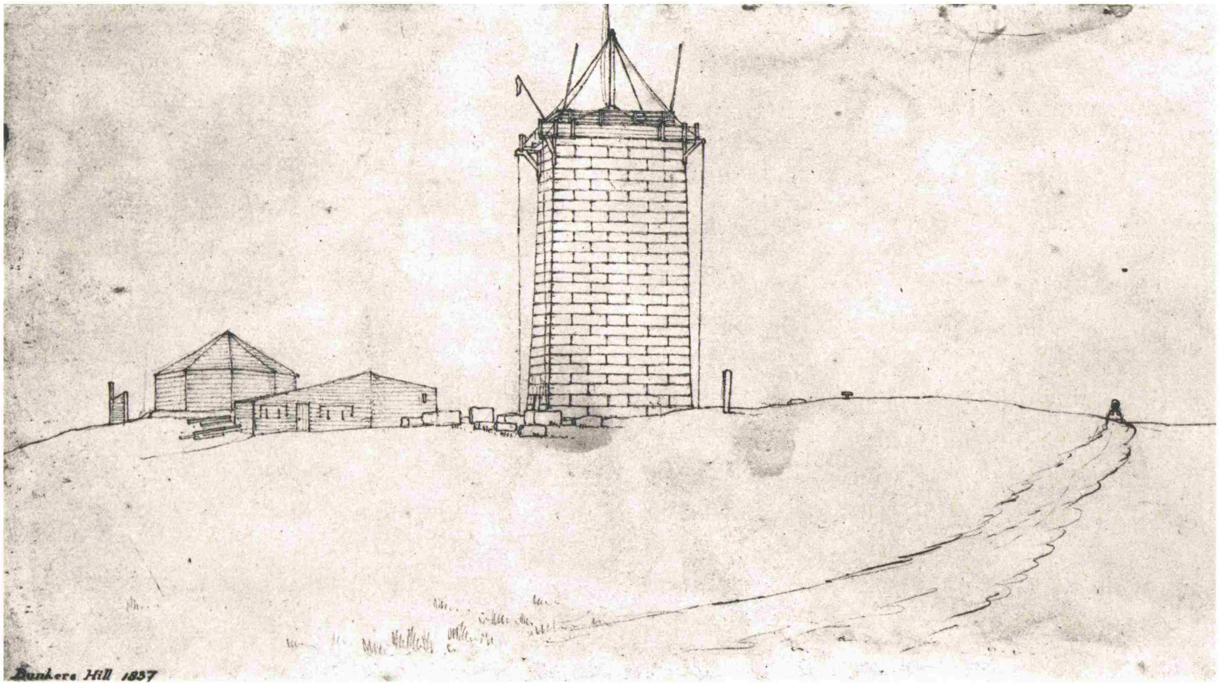
[3] William W. Campbell, *Life and Writings of DeWitt Clinton* (New York: Baker and Scribner, 1849).

As shown in the table of dimensions which follows, the monument, almost exactly, is built to the dimensions of the Baldwin Report,[4] which was influenced by the Greenough model.

Height, above ground	220 ft.
Sides of monument, at ground level	30 ft.
Sides of monument, at base of apex	15 ft.
Height of apex	12 ft.
Minimum wall thickness, at base	6 ft.
Diameter of circular interior, at base	18 ft.
Height of masonry elements:	
78 main courses, each with height of	2 ft., 8 in.
5 courses in apex, each with height of	1 ft., 8 in.
Height of capstone	3 ft., 6 in.

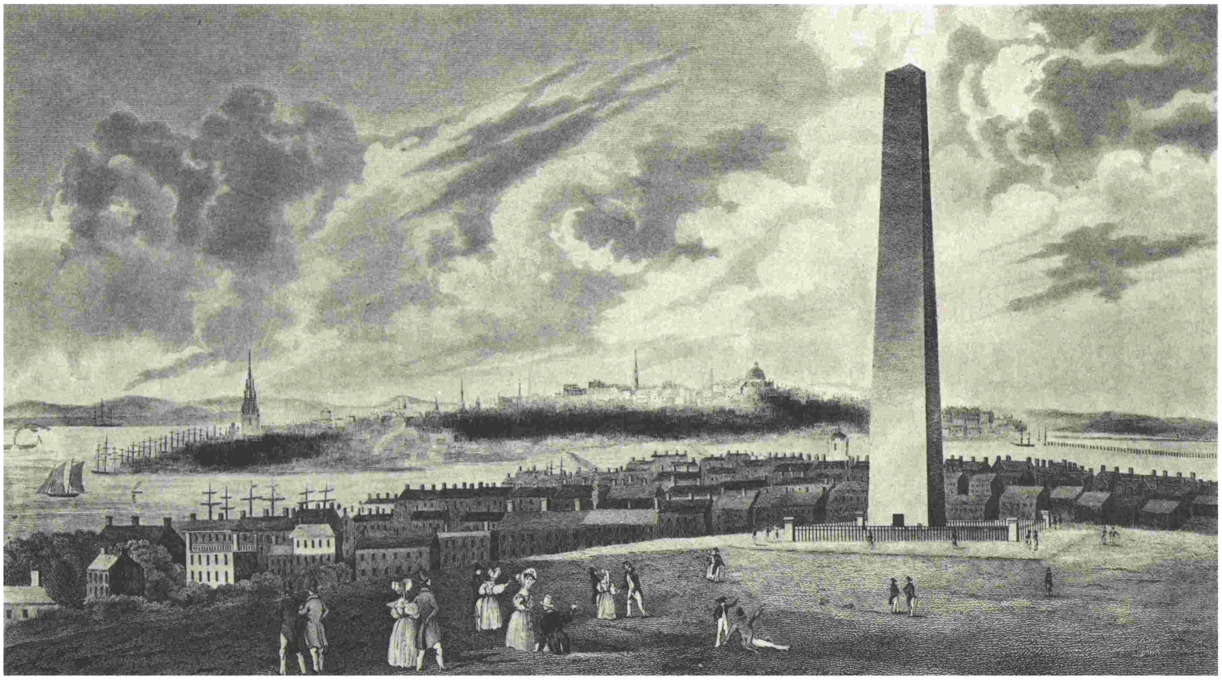
[4] Measurements given above have been taken from: *Baldwin Report* (1825)—Loammi Baldwin; *Plans and Sections of the Obelisk on Bunker's Hill* (1843)—Solomon Willard; *History of the Battle of Bunker's (Breed's) Hill* (1875)—George E. Ellis.

Data from old records do not always check modern measurements. For example, a modern reference gives the height of the monument as 221 feet.



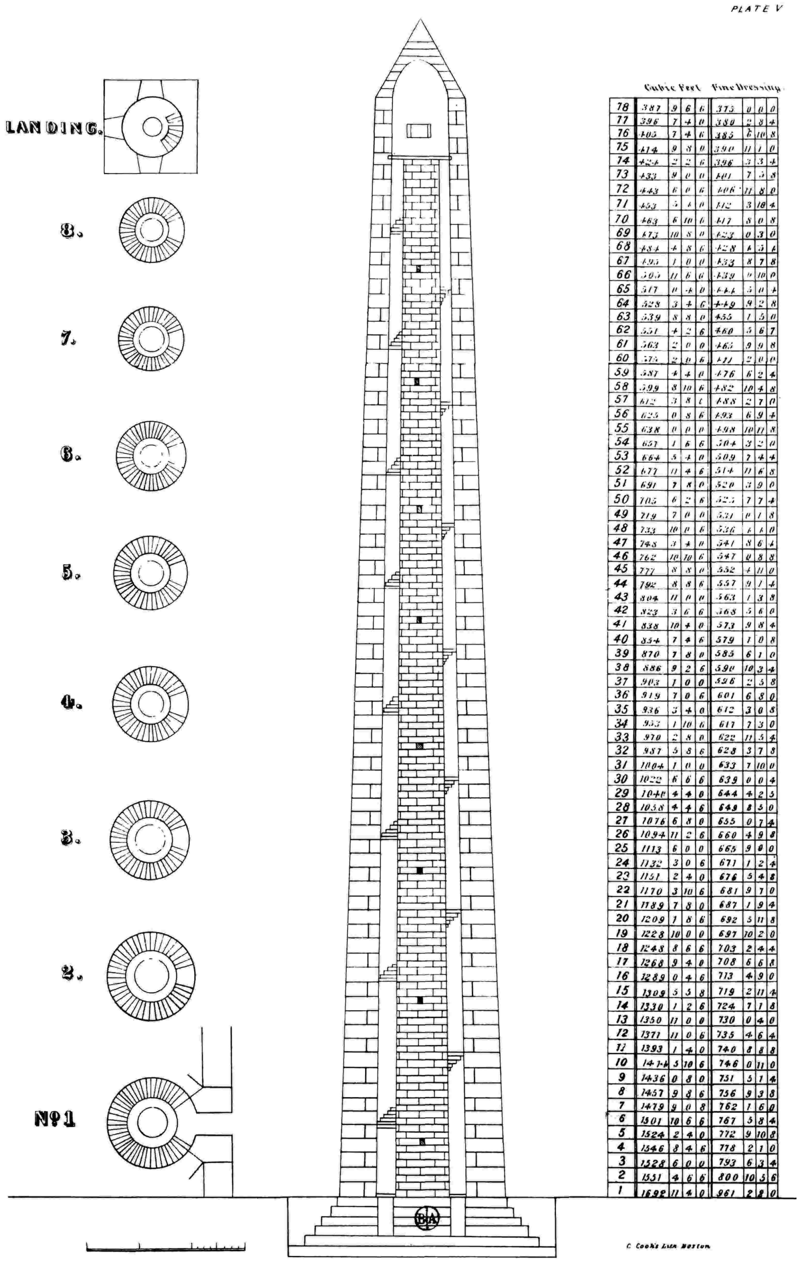
Bunker Hill Monument under construction in 1837

Reproduced from a hitherto unpublished drawing in the Boston Athenæum



Bunker Hill Monument as proposed

Reproduced from an 1834 certificate of membership of the Bunker Hill Monument Association in the Boston Athenæum



Section of Bunker Hill Monument

Reproduced from Solomon Willard, *Plans and Sections of the Obelisk on Bunker's Hill*

(Boston, 1843), Plate V

As described in the Baldwin Report, the circular winding staircase is composed of granite steps, starting with a width of about four feet and narrowing as the ascent is made. Baldwin called for "places of repose" (landings) at intervals. Modern architects call the part around which a circular staircase winds, the "newel." Baldwin's newel is a hallow wall, 10 feet in diameter at the base, about two feet thick.

Thus, the monument was designed by an engineer, not an architect. Baldwin violated a common rule for the proportions of ancient Egyptian obelisks, that the pyramidion should be as high as the base is wide, which is one reason why the Washington Monument is so beautiful. One regrets that architect Willard, who picked up where Baldwin left off, did not see fit to modify the Baldwin lines. There seems never to have been any question as to the monument's material: granite, the native New England stone. Although we admire the Bunker Hill Monument for its somber strength, it cannot be called a structure of beauty, as is the lighter-tinted and finer-textured marble Washington Monument, with its sharper apex.

We can also speculate on why Baldwin made the monument wholly of granite. At today's prices, the circular inner surface of the shaft and the circular chimney, or newel, around which the stone staircase winds, would be of tremendous cost. The dressing of the stone for a square inner area would be much cheaper.

Before criticizing Baldwin on his ponderous stair design, which could be replaced by a light, modern fire escape, we should look at the status of the tiny American iron industry of his day. The ironmasters were recovering from the decline of activity in the War of 1812, during which they had lost their British market. Baldwin would know that certain early railroad promoters estimated that granite tracks mounted with iron plates would be less expensive than the English-rolled rails, which the Americans could not produce. With masons in Massachusetts receiving about $0.18 an hour, granite was considered cheaper than iron. Baldwin therefore designed his stairway of granite, with a massive granite chimney "newel" to support the inner ends of the treads. Long before the monument was completed, however, a square staircase of either cast iron or wrought iron could have been produced, economically, by American ironmasters. It was then too late to make the change, however.

At about the time of the completion of the monument the first mechanical elevator was exhibited, but there was no room for an elevator at Bunker Hill—the newel was in the way. To climb a few score steps would be an easy task to our sturdy forebears, and to say that one has "climbed the Bunker Hill Monument" is a boast that hundreds of thousands of tourists to Boston have been proud to make for over 100 years. Baldwin may have been right again, as he usually was.

Baldwin specified that the monument should be square with the compass, a common Egyptian practice. As built, however, it is oriented to fit the redoubt (southeast corner) of the battle fortification. Structurally, Baldwin designed a sound foundation, 12 feet deep, built of six courses of stone with no small rubble that might deteriorate through the years. He specified that the starting level of the base of the monument should be established at the best elevation to avoid an uneconomical distribution of the excavated earth; today we would say that he balanced cut and fill.

The modern building contractor finds the estimate that went with the report both practical and quaint. He will find that the digging of the pit for the foundation of the monument was figured in "squares," at $2.00 each; and since a square meant eight cubic yards, hand excavation was therefore priced at $0.25 per cubic yard. This price must have included the expense of shoring; also the pumping that such a deep pit would require. Baldwin proposed to dig a deep well on the site (today called a test pit), which would not only indicate the adequacy of the soil, but would also furnish water for construction purposes. Much water would be needed to mix the lime and sand mortar for the monument as well as for the Roman cement, for which the estimated 100 casks were figured at $7.00 each.[5]

[5] "The use of natural cement was introduced by Mr. Parker, who first discovered the properties of the cement-stone in the Isle of Sheppy, and took out a patent for the sale of it in 1796, under the name of 'Roman Cement.'"—Edward Dobson, *Rudiments of the Art of Building* (London: John Weale, 1854).

Masonry was then estimated in "perches," and by a little arithmetic, the modern contractor will learn that a perch was then equal to 25 cubic feet, or nearly a cubic yard. The 784 perches of masonry for the foundation were priced at $10 per perch, including "stones, hammering, mortar, laying, etc."

The report of Baldwin contains no computations on the structural stability of the monument. If the modern structural designer wishes to investigate how near the safe limit the monument has been tested by Boston's occasional hurricane winds, he has available the major dimensions given in the Baldwin Report, and the drawings of Willard's classic *Plans and Sections of the Obelisk* from which to make this simple computation.

Such computation shows that the monument is so heavy that a hurricane wind has an almost imperceptible effect on its stability. When it is subjected to a 100-mile-per-hour wind, the resultant force is displaced only a fraction of a foot from the center of the 50-foot-wide foundation. The maximum load on the soil is about five tons per square foot—a safe bearing load on "the bed of clay and gravel which composes the soil of the Hill" as described in an old account. The same account speaks of "great pains having been used in loosening the earth, and in puddling and ramming the stones." Surely, our construction ancestors would not have purposely disturbed the underlying soil, in an attempt to improve upon the natural bearing strength of one of the firmest of foundations: glacial hardpan. Like any good builder, they were undoubtedly merely puddling with water the earth backfill around the completed foundation.

Baldwin knew that granite would not deteriorate when exposed to the alternately hot and cold temperatures of Boston. Half a century later, the engineers who transported an Egyptian obelisk (one of the Cleopatra's Needles) to Central Park, New York, learned that the lovely textured syenitic granite of the Nile Valley was markedly inferior to New England granite in weather resistance, although it had kept its surface intact for centuries in the mild climate of Egypt. To protect Cleopatra's Needle in New York, a paraffin coating was found necessary.

Baldwin soon resigned from the building committee, partly because of the press of other work, but largely in protest against a clause which made its members, all of whom freely donated their services, financially responsible for the estimate. Promptly after accepting his resignation, the directors revised this clause. In reviewing the quaint old methods, the question arises: Would modern estimates be more accurate if the consulting architects and engineers had to pay for overruns?

Transient Cornerstone

On 17 June 1825, the cornerstone of the monument was laid with impressive ceremonies. As the colorful procession marched up Bunker Hill to the stirring rendition of “Yankee Doodle” by the drummer of Colonel William Prescott’s regiment, who, 50 years before, had been in the battle, the rear of the procession was just starting from distant Boston Common. The little Boston of over a century ago was crowded with visitors who had come from places as remote as South Carolina by stagecoach, sailing vessel, or on foot, to hear the great speech of Daniel Webster, President of the Bunker Hill Monument Association, and America’s first orator of the day. Years earlier, Chaplain Joseph Thaxter had paid the last offices to dying soldiers in the battle; now, he invoked God’s blessing on the young American republic, as 40 veterans of the battle sat in a place of honor.

The most important visitor, of course, was General Lafayette, who, as a good Mason, spread the mortar on the stone when it was laid by Most Worshipful Grand Master of the Grand Lodge of Massachusetts, John Abbot. As the battle’s only monument up to this date had been erected by the Masons, it was considered appropriate that the permanent monument should have its cornerstone laid with the Masonic ceremony. A little later, this procedure was sharply criticized during the Antimasonic period, which occurred before the monument was finished.[6]

[6] Joseph Warren, the outstanding hero of the battle, was Grand Master of Freemasons for North America.

Many of the spectators knew that the cornerstone records would later have to be moved, for the plans of the monument were hardly started. Now, the box with its old newspapers, Continental currency, and other data is within a stone at the monument's northeast corner, and the original cornerstone stands in the center of the foundation.

With his usual generosity, Daniel Webster presented the copyright of his famous speech to the Bunker Hill Monument Association. The copyright was sold for $600, which was the second largest single contribution up to that date.

The Leading Character

Solomon Willard, architect and superintendent of the Bunker Hill Monument, developed the methods for the quarrying, dressing, transporting, and erecting the huge stones of the monument that started granite on its way to becoming a principal material for massive structures in America for half a century, until reinforced concrete took over. (Today, granite is used extensively as a protective facing for concrete, for highway curbing, and for memorials.)

It is impressive to note the universal respect for the integrity and ability of this early American architect which all the records of the monument stress. In the drama of the building of the Bunker Hill Monument, he played the leading part, and his character resembled the sturdy structure which he designed in detail and erected. During his 18 years of service in the construction of his masterpiece, the Bunker Hill Monument, he would accept no recompense except for his expenses, deeming it his duty to work without pay on such a patriotic venture. He was also a substantial contributor to the building fund.

A self-educated man, who had learned architecture with sufficient thoroughness to become a teacher in the subject, he had also become proficient in the various sciences. Starting as a carpenter, Willard had proved both his craftsmanship and artistry by becoming an adept carver of ships' figureheads and models, including a model of the Capitol at Washington.

At the time the monument was begun, Willard was one of the leading architects of Boston. Typical of an architect's versatility, he had played an important part in the change from the heating of buildings by wood-burning fireplaces and Franklin stoves, to hot-air furnaces, using either wood or coal. As an expert in furnace heat, he was called in for advice in the design of the heating system of America's most important building, when the President demanded that the national Capitol should have adequate heat.

Appointed in October, 1825, to the combined position of architect and superintendent, Willard spent the following winter on the plans, models, and computations required to develop the construction details, from the over-all dimensions of the Baldwin Report. During these preliminary steps, Willard experimented with a promising machine for dressing the stones. The selection of the Bunker Hill Quarry in Quincy, Mass., was made after Willard had made a careful search for suitable stone, in which he was said to have walked 300 miles. The right to quarry, at Quincy, sufficient granite for the monument was purchased for $325. Part of the amount to be provided by the Commonwealth of Massachusetts was to have been supplied by the cost of the dressing of the stone (then called "hammering") by the convicts of nearby Charlestown State Prison. The convicts, however, were obviously not sufficiently independent to work on this shrine of independence, so this procedure was not adopted.

Up-To-Date Quarry (Circa 1825–1843)

From various old American and English records of masonry construction, it is possible to construct an account of how the stones for the Bunker Hill Monument must have been quarried and dressed. The old names are used for the tools and methods, and the modern mason will find many of these old descriptions quite familiar.

The hornblende granite of the Quincy region was (and is) of very uniform texture and varies only in color, from gray to dark gray. In Quincy, Willard would find that both "sheet" and "boulder" quarry formations occurred: the joints in the ledge of the sheet areas making the granite appear as if stratified, and hence more easily removed; but the huge, rounded boulders in the other areas, measuring up to 40 feet across, had no joints. Rows of holes were drilled by hand (at least 25 years would elapse before practical power-rock drills became available) and large blocks loosened from the ledge or boulder, probably by wedges, possibly by light blasts of gunpowder. At this stage the quarried block was called “quarry-pitched.” Stone of the smaller size for the monument was split from these blocks along lines of holes in which wedges were driven. These were probably of the plug-and-feather type, in which an iron wedge with an acute angle is fitted between two semicircular iron feathers, which taper in the opposite direction to that of the wedge, and thus fit the hole drilled in the stone, nicely. Granite has no cleavage planes, like slate; but a routine of smart taps on the plugs, back and forth along the line, soon splits the stone along a fairly smooth face. Two lewises (an ancient device), attached at about the quarter points of the top of the stone, were used to lift it. Three members make the lewis: a flat center bar with an eye at the top, the center bar being flanked by two wedge-shaped side pieces which are thicker at the bottom of the hole, and these also have eyes at the top. The wedges are inserted first, then the center bar slipped between; thereafter any lifting pull on the three bars is bound to expand the lewis to fill the hole and lift the stone, for the hole is drilled wider at the bottom than at the top.

With Solomon Willard’s well-rendered isometric drawing of each stone for a guide, the stonecutter dressed it, first selecting the best face for the “bed” (bottom) and hammered it to a plane surface, determined by shallow channels (chisel drafts) cut diagonally across the stone.

From this surface, the stonecutter laid out the other faces, including the “build” (top), by his good mason’s square or template. The texture of the visible face was “tooled,” that is, the marks of the chisel remained visible. Quincy granite is a quality product, taking a high polish, but the builders of the Bunker Hill Monument desired no polish on their monument. Today, the surface of the monument shows faint, well-weathered lines, like those produced by the modern bushhammer, which has a head made of several thin steel plates bolted together, each sharpened to a cutting edge. In England during the period, flat iron bars with rough edges were in use to saw softer stone than granite, and at Quincy, Willard experimented with dressing machines. The conclusion may be drawn, however, that the stones which we now see on the monument were undoubtedly shaped to their present dimensions by hand.

Today, 110 years after its capstone was put in place, the Bunker Hill Monument stands as an impressive testimonial to the conservative judgment of its designer, Loammi Baldwin, and the painstaking fidelity of the man who supervised its construction, Solomon Willard. An engineer familiar with its maintenance states that there is no evidence of settlement, and that a check by surveyor’s transit revealed no signs of misalignment. Its joints occasionally need pointing, the last pointing being performed about 20 years ago. Various iron or steel members of the observation chamber have had to be replaced. Its lightning rod has been in place for many years, but there is no readily available record to check whether the monument has ever been struck by lightning. With their empirical methods of design and their crude, mostly hand-operated, construction apparatus, our forebears built a sturdy structure, which, barring an earthquake, should last for centuries.

The Granite Railway

On 7 October 1826, the first railroad in America started operation. This was the horse-operated Granite Railway, built to transport the stones for the Bunker Hill Monument from the quarry in Quincy down to the Neponset River, a distance of nearly three miles. The track and cars of the railroad had been designed and built by a young engineer of 28, Gridley Bryant, whose Granite Railway project started him on a long career of achievement in the invention of equipment that played a major part in the rapid and successful development of the American railroad system.

Ample precedent for the Granite Railway existed in England, where, since the reign of Charles II, wooden tracks, sometimes armored with iron plates, had been used as runways for coal cars from the pits to the nearest waterway. Within five years of the start of the Granite Railway, similar systems are recorded in the states of Pennsylvania, South Carolina, New York, and Maryland. At first, the motive power for these lines was supplied by gravity, stationary engines, or horses, but soon tiny steam locomotives were tried. Thus, in the year 1829, Peter Cooper built the famous *Tom Thumb*, a successful locomotive which used rifle barrels for flues. In the same year the *Stourbridge Lion*, “the first locomotive that ever turned a driving wheel on a railroad on the Western Continent,”[7] was brought by sailing vessel from England and started operation in Pennsylvania. The American steam railroad system was thus well under way by the time the lower courses of the monument were being raised.

[7] *Annual Report of the Smithsonian Institution*, 1889.

Bryant later described his railroad as having stone sleepers laid across the track, 8 feet apart. Upon these, were placed wooden rails, 6 inches thick and 12 inches high (replaced by stone within a few years). Spiked on top of these were iron plates, 3 inches wide by ¼ inch thick. However, at road crossings, stone rails were used, with 4-inch by ½-inch iron plates bolted on top. This "permanent" construction was also used on the double-track, inclined plane at the quarry. (Well-preserved vestiges of this "permanent" construction are visible today at the rise to the Bunker Hill Quarry.) Here, an endless chain allowed the loaded, descending cars to pull up the empty ascending ones.

The standard gauge of American railroads is now 4 feet, 8½ inches, measured between railheads, a standard adopted after many years of confusion before the present gauge dimension was adopted. Although Bryant described his track gauge as 5 feet, this dimension was measured between the "bearing points" of the wheels on the tracks. If the bearing points are assumed to be the center of the treads of the wheels, his gauge is found to match closely the present standard gauge. This track gauge agrees with that adopted by the famous English railroad engineer, George Stephenson, at about the same time, after he had measured scores of carts used by his farmer neighbors. Possibly, both Stephenson and Bryant knew that their selected gauge had a very early beginning; for some historians suggest that the English carts were originally made to fit the ruts cut in the roads of Britain by the Roman chariots, many centuries earlier, during the Roman occupation of Britain.

On the day when the railroad started operation, 16 tons of granite from the Bunker Hill Quarry, and loaded on three "wagons," were easily pulled by one horse, once started. Bryant's first car had flanged wheels, 6½ feet in diameter, from the axles of which a platform was hung to carry the granite. This platform was lowered to receive the load and then raised by an ingenious gearing device.

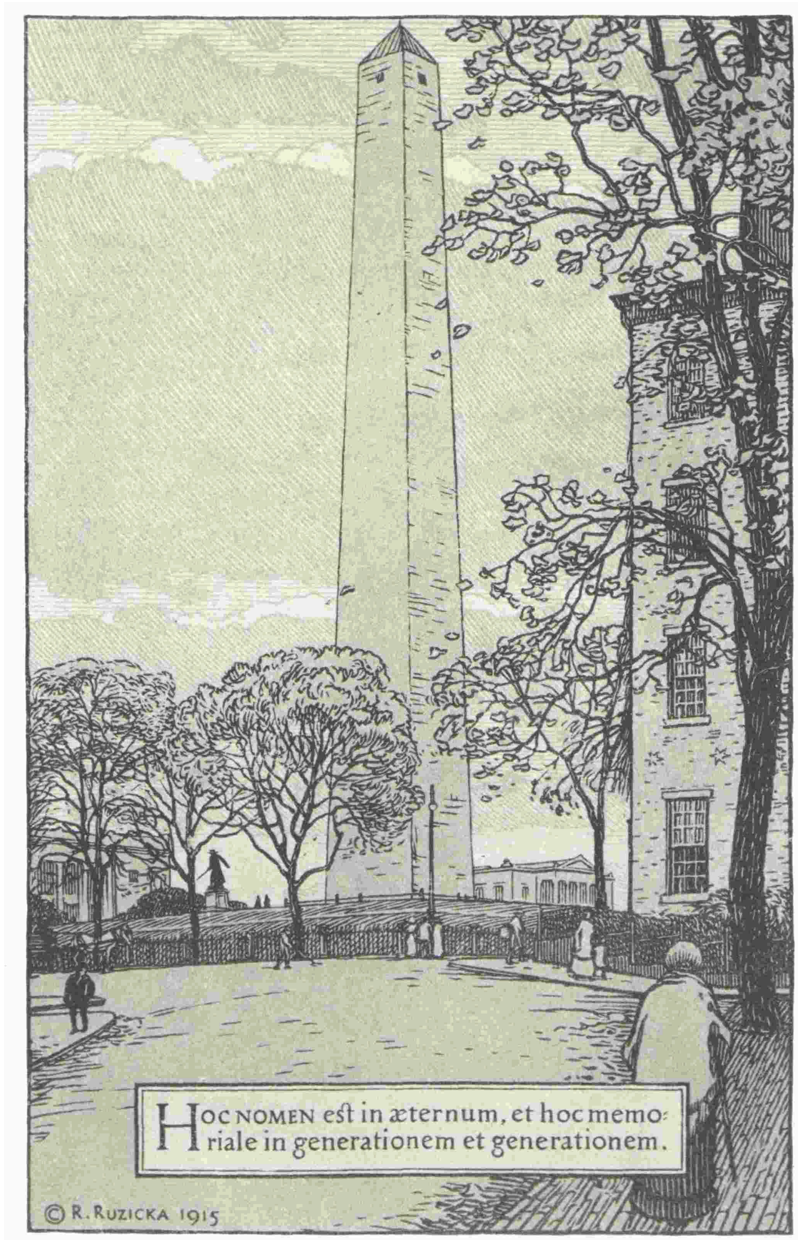
Naturally, Bryant based the design of his early railroad cars upon the construction of the horse-drawn wagons of his day. Like the wagons, his cars had to be flexible if they were to keep on the track when passing over the two curves of the otherwise straight Granite Railway. In his description of another of his cars appear the road wagon terms—bolsters, truck, and center kingpin, to allow a swiveling motion. Rigidly bolted to cross timbers beneath the truck were two iron axletrees, on which revolved cast-iron wheels. (Some time would elapse in railroad progress before the wheels would be fixed to, and revolve with, the axles in journals.)

In early American railroad development Bryant is credited with the invention of the eight-wheel car, the turntable, switch, turnout, and many other improvements. In 1832, he had invented and used in the building of the United States Bank at Boston, his portable derrick, "used in every city and village in the country wherever there was a stone building to erect." Others profited from Bryant's amazing ingenuity. Although the Supreme Court of the United States decided in his favor in his most important invention, the eight-wheeled car, he did not collect, and he died poor.[8]

[8] For more data on the Granite Railway and Gridley Bryant, *see*: Charles B. Stuart, *Civil and Military Engineers of America* (New York: D. Van Nostrand Company, Inc., 1871); and *The First Railroad in America* (Boston: Privately printed for Granite Railway Company, 1926).

In the fine saga of the Bunker Hill Monument, the Granite Railway plays a prominent part. The demand of the monument for granite definitely inspired Bryant to conceive the idea of America's first railroad, and to design pioneer equipment that contributed hugely to the subsequent progress of America's great railroad system. The accurate account of the building of the monument, however, has to record the fact that the railroad was not so great a benefit as anticipated. In the short distance of 12 miles there was too much loading and unloading. Willard freely expressed his annoyance at these hindrances. That he took action is indicated in the following quotation from an apparently authentic source: “The stone used for the foundation and for the first forty feet of the structure (the monument) was transported from the quarry on a railroad to the wharf in Quincy (actually located in Milton) where it was put into flat-bottomed boats, towed by steam-power to the wharf in Charlestown, and then raised to the Hill by teams moving upon an inclined plane. The repeated transfer of the stones, necessary in this mode of conveyance, being attended with delay, liability to accident, and a defacing of the blocks, was abandoned after the fortieth foot was laid, and the materials were transported by teams directly from the quarry to the hill.”[9] This account fails to tell how the teams got up and down the steep hill at the quarry: the 84-foot rise at an angle of 15 degrees. Clever Bryant must have used his endless chain to drag the empty teams up, and to brake the loaded ones down.

[9] George E. Ellis, *History of the Battle of Bunker’s (Breed’s) Hill* (Boston: Lockwood, Brooks and Company, 1875).



HOC NOMEN est in æternum, et hoc memoriale in generationem et generationem.

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Bunker Hill Monument in 1915

Reproduced from a wood engraving by Rudolph Ruzicka in the Boston Athenæum

Judah Touro

Reproduced from a portrait in the Redwood Library,
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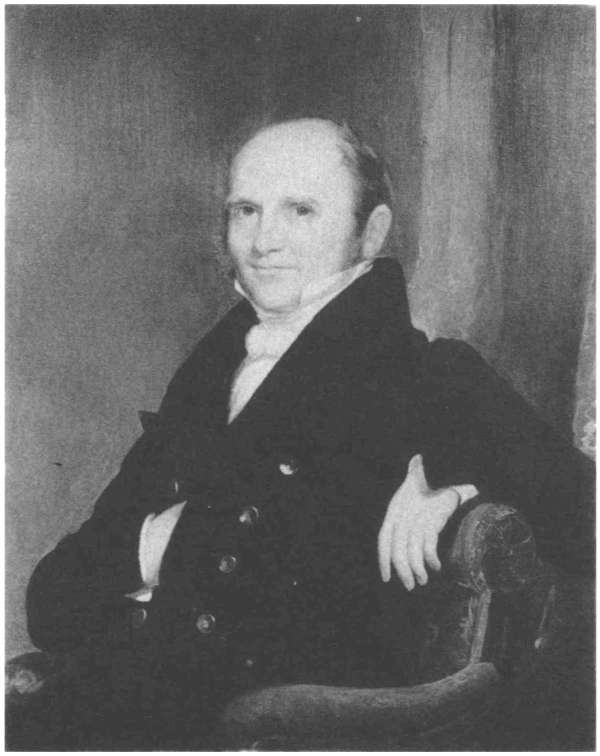

Amos Lawrence

Reproduced from a portrait by Chester Harding
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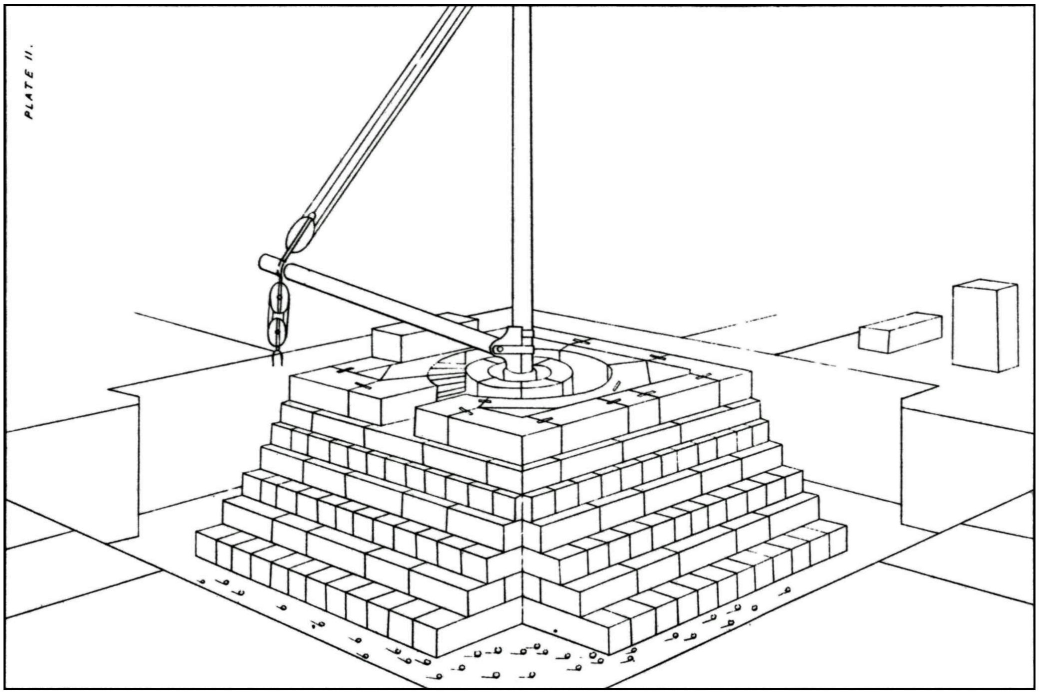
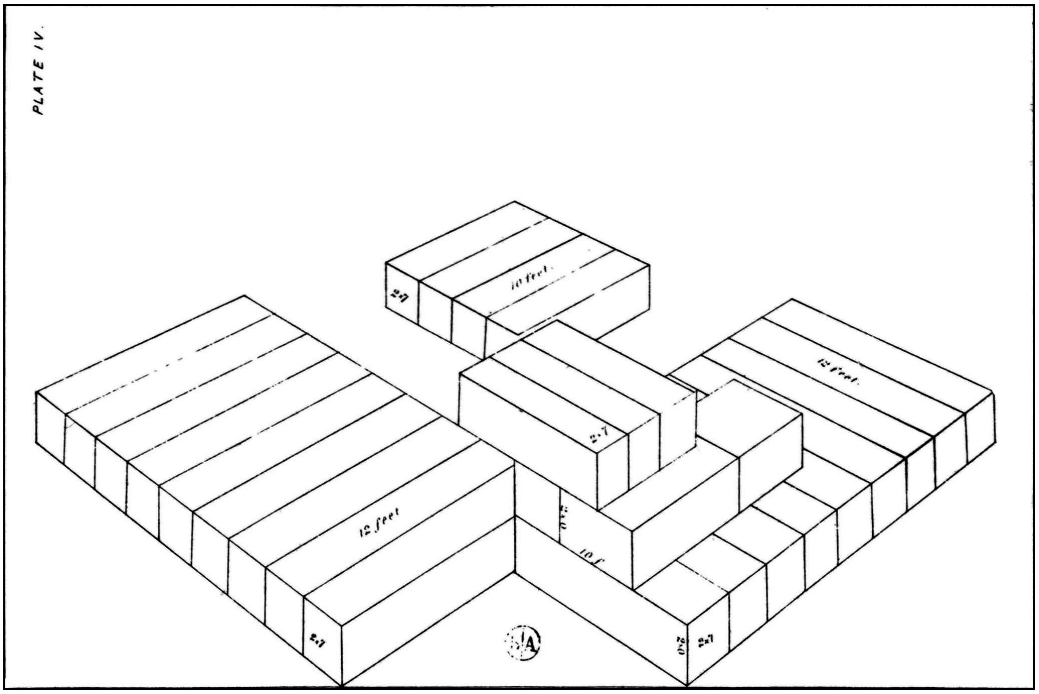


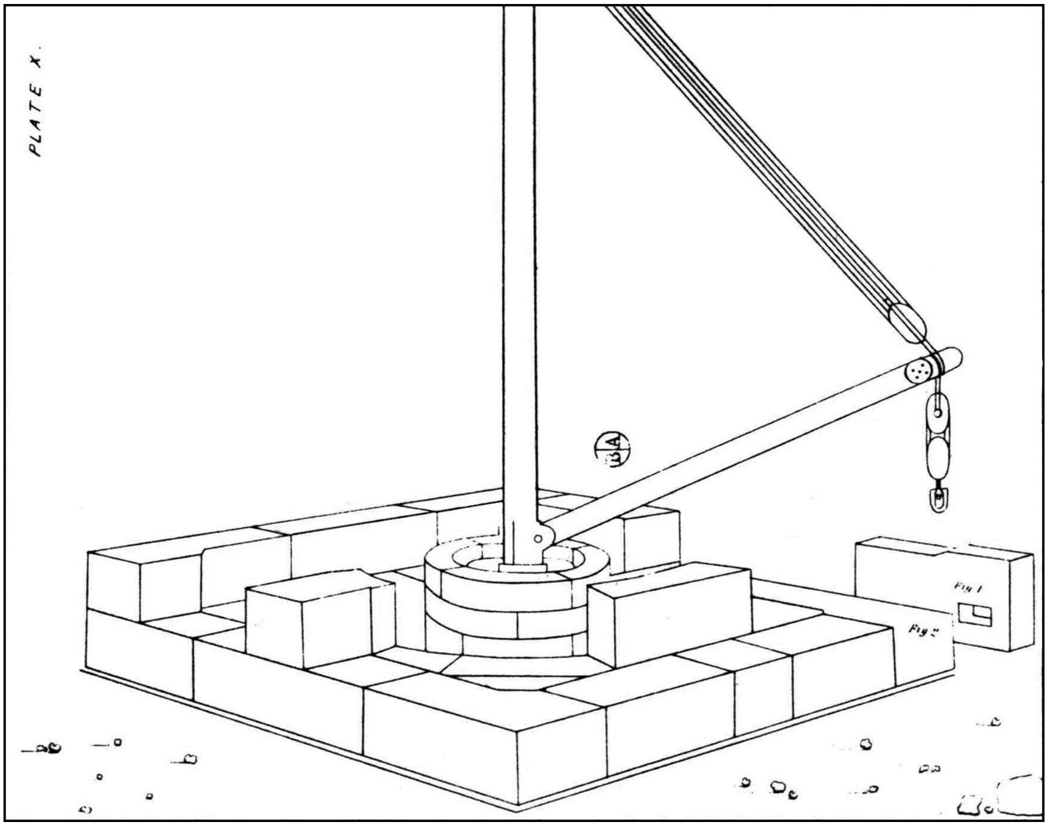
Foundation of Bunker Hill Monument

Reproduced from Willard, *Plans and Sections*, Plate II



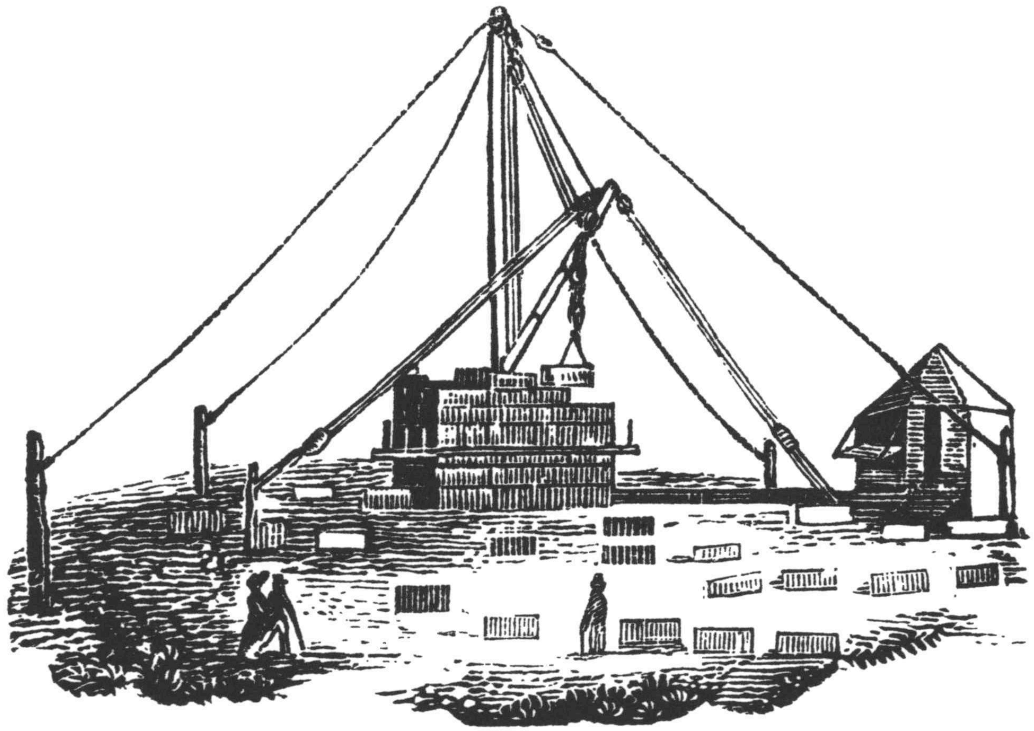
Construction of Bunker Hill Monument

Reproduced from Willard, *Plans and Sections*, Plate IV



Construction of Bunker Hill Monument

Reproduced from Willard, *Plans and Sections*, Plate X



Bunker Hill Monument in 1830

Reproduced from C. H. Snow, *A Geography of Boston* (Boston, 1830)

Beacon for Mariners

In the noisy grogshops on the streets leading to the Boston waterfront, in the sail lofts on what is now Commercial Street, and at the tall desks of the counting rooms of State Street, those who got their living from the sea eagerly discussed the progress of the monument in Charlestown. It was to be their beacon, and when the many frigates, packets, sloops, and schooners had safely passed the danger spots of the lower harbor, the monument would welcome them to the busy inner port of Boston, then much livelier than it is today. But progress proved to be slow. Naturally the stones broken from the Quincy ledges and boulders were not always of the dimensions planned by Willard for the lower courses; many were of sizes needed for the upper courses. Economical Willard dressed the stones as they came out; setting aside those which could not be erected for some time; and the piles of such stones grew larger at Quincy and on the ground about the monument, while the monument itself rose at a snail's pace. A more spectacular progress was needed for a project that was started on a shoestring, and depended on more and still more public contributions. The building fund dwindled to such a low sum that in February, 1829, work had to be suspended for lack of funds to pay the wages of quarrymen, stonecutters, derrickmen, blacksmiths, and teamsters, and the cost of the good hay for the hard-working horses and oxen of the project. But 14 courses had been laid—to a height of 37 feet, 4 inches. The sailors were disappointed, and a poetess said:

> But where's the pile they said would rise,
> Throwing its shadows o'er the wave,—
> Lifting its forehead to the skies—
> A Beacon far o'er land and sea,
> Signal and Seal of Liberty.

A lottery to secure more building funds was next proposed. It was not unusual to allow lotteries in this period—churches, turnpikes, bridges, and even Harvard College—had received such grants. Public sentiment in Massachusetts, however, was beginning to consider lotteries a vicious practice and the director of the Bunker Hill Monument Association voted against one.

At this time, Amos Lawrence was a member of the building committee—a wealthy philanthropist of Boston, whose religion seemed to be fixed on two ideals, one of which was his charities. Unlike that of a few good ministers of the time, who had preached against giving funds for the monument because they felt their various charities should come first, Lawrence deemed the completion of the Bunker Hill Monument of first importance. This project became his other obsession. He enlisted the aid of the Massachusetts Charitable Mechanics Association in the campaign for funds. Started in 1795 by Paul Revere, and others, to promote a better understanding between master mechanics and their apprentices, this society had become influential; its membership embraced mechanics, manufacturers, and such honorary members as Ex-President Adams, Daniel

Webster, and Edward Everett. Amos Lawrence had picked upon a well-managed organization for assistance; its executives were shrewd financiers and they knew how to get things done. The president of the Massachusetts Charitable Mechanics Association became in perpetuity the first vice-president of the Bunker Hill Monument Association (today he is still so listed). Thereafter, the Mechanics Association took an active part in the promotion and construction of the monument. It made a careful estimate of the cost to complete the monument and, much to Solomon Willard's disgust, raised his allowance for contingencies. Both associations decided to be satisfied when the monument had reached the height of 159 feet, 6 inches—about two-thirds of the height previously determined upon.

Work was resumed on 17 June 1834, and continued until funds again gave out, when the monument was 32 courses high, 85 feet; now imposing enough for quite the good mariners of busy Boston harbor to take notice of. It was the year 1835, and the country was headed for a severe financial depression—a bad sign for those who sought contributions for any but the most practical of objectives. In this emergency, the women of New England again became active in the raising of funds.

In the summer of 1840, a common greeting of the women of Massachusetts was: "What are you doing for the Fair?" Those who knit stockings, crocheted in worsted of various colors, who were skilled in embroidered work, or who were merely good at plain sewing strove industriously to get ready for the Fair that was to earn money for the Bunker Hill Monument. Ten years earlier, the women of New England had made a noble effort to secure funds for the monument, but as the contributions from females and children had been limited to a maximum of $1.00 each, the total was small. Now, although the maximum sum ever raised at a Boston fair was $3,000, they felt that a sizable sum could be realized in a fair in Quincy Hall, near Faneuil Hall. Despite the criticism that "women were stepping out of their sphere," Sarah J. Hale, the leading spirit in this remarkable effort, persisted.[10] Quincy Hall was 382 feet long by 47 feet wide, and it was crowded with the 43 tables of things to sell, when the seven-day fair started in September, 1840. A Whig Convention, in this year of a presidential election, undoubtedly helped to increase the attendance at this very successful fair. The price of admission on the first day was double that of the remaining days, which was $0.25.

[10] She was editor of *Godey's Lady's Book* for 30 years, and is credited with promoting the establishment of the last Thursday in November as Thanksgiving Day.

The success of the Quincy Fair was phenomenal. The net sum of $30,035.53 was realized and turned over to the directors of the Bunker Hill Monument Association toward the completion of the Bunker Hill Monument. This amount was nearly one quarter its total cost. The Yankee ladies did not know that two contributions of $10,000 each, with several smaller donations, were available by now. The gift of $10,000 by Judah Touro was peculiarly heartening, as an example of the expression of the gratitude of the son of an immigrant to the country of his adoption. The father of Touro had been rabbi of a synagogue in Holland. The younger Touro was born in Newport, R. I., in 1776; he had sailed to New Orleans with an assortment of New England commodities and had made money in their sale. A soldier in the Battle of New Orleans, in the War of 1812, he had been given up for dead in the battle. He had become a millionaire and, learning of the proposed gift of $10,000 by Amos Lawrence, toward the completion of the monument, Touro had matched it. Thus, in the year 1840, the success of the Bunker Hill Monument was assured. It could now be built to the full height planned by Baldwin and Willard—about 220 feet.

The Riggers

Contractor James Sullivan Savage would have no trouble finding good men for the ticklish job of raising and setting the heavy stones of the higher courses of the monument; able sailors, who would take a shore job for a change. Maritime Boston was full of these good riggers, who were used to dizzy heights, and to whom the half hitches, square knots, guys, slings, and tag lines would be easy. Up to this time the monument had been built by day labor, not by contract. Now, Savage had taken a contract to finish the monument for $43,800, from the elevation of 85 feet to the top. He was well trained in masonry, for he had worked on the job since the start under Willard, whose rigid ideas would not let him take a contract himself for profit on such a patriotic project, but who agreed to superintend the work of Savage to the finish. Savage had the traits of a good contractor—energy, resourcefulness, honesty—and the sense that knew how each detail must be executed toward the end of producing a job to be proud of.

Savage replaced the one-horse capstan of the hoist by a six-horsepower steam engine, an innovation that speeded up progress. The steam engine as a prime mover in land and water transportation had become well established, and its use to drive textile machinery had proved successful. Steam power in the construction industry, however, was a novelty. Shouts and wigwag signals from the setting gang at the top to the engineer on the ground were replaced by a bell-wire signaling system. This must have been a pull bell, for many years would pass before electric bells came into common use.[11]

[11] Joseph Henry had developed the electromagnet at about the time of the laying of the lower courses of the monument, and, a few months after its dedication, Morse would operate the first telegraph line between Washington and Baltimore, but the transmission of electric currents by insulated wires even for a few score feet was still too new to receive serious attention on a construction job.

As its lighter stones would be easier to handle, the granite inner cone (newel) around which the stairs wind was erected a few courses ahead of the walls of the monument. It thus served as a support for the derrick which raised the heavier wall stones. Through apertures in the hollow walls of the newel, a heavy beam (wood?) was passed, upon which the derrick was set.

It is interesting to compare our modern hoisting derrick with the apparatus used to raise the stones for the monument. The derrick of today consists of a guyed, vertical mast, an adjustable boom hinged to the base of the mast, with boom falls, and hoist falls, each with their cables and pulleys, or blocks. At the base of the mast, a bull ring serves to turn the mast by power. The whole combination is called the derrick. When we get accustomed to the old English or American custom of calling the mast the post or derrick, and the boom either the gaff or derrick, a little study enables us to comprehend how the monument was built.

The lower courses were raised by the “Holmes Hoisting Apparatus,” designed by a practical seaman of Boston. This device could command a circle 100 feet in diameter. Except that it had no bull ring to turn the mast, it appears much like the derrick of today. With steam power available for the upper courses, Savage seems to have modified the boom of Holmes to serve as a nearly horizontal “lever,” on which a “wheel carriage” drew the stone inward, to its desired position for placement. In other words, apparently, the boom became today’s monorail. A somewhat obscure, English description of the means used to hoist masonry 100 years ago, tells of two devices. One was a “movable derrick crane,” with a vertical post, supported by two timber backstays, and a movable hinged “jib or derrick,” which could be today’s boom. This assembly, of course, corresponded to today’s stiff-leg derrick, in which the back guys are replaced by timber members. The other English device for raising stones was practically exactly like today’s traveling crane, and that was the name it went by in England, 100 years ago.

Our construction forebears of over a century ago had to use ropes and chains for all purposes; there were no wire ropes. About the time Savage set his first stone, John A. Roebling was making the first American wire rope cable, in a largely outdoor plant located on a level meadow on his farm in Saxonburg, Pa. Wire rope had real advantages in construction work, because of its superior strength and its much less stretch under load. A crude sketch, dated 1837, shows that the derrick for the monument was guyed by chains, which attached to the top of the mast and passed over timber brackets at the staging level, and thence vertically down to weights at the ground. In his long length, the stretching and shrinking of a rope under rain, load, and temperature changes would be difficult to control.

Every four courses of the wall stones (about 11 feet) the derrick was raised, perhaps by unshipping the boom and using it as a lesser mast to raise the mast proper. Square timber sticks were then beginning to be used for staging, instead of the round trunks of small trees and saplings previously used. A sketch of the monument shows a squared timber stage for pointing the joints. Such a stage could be much more easily erected, and was more reliable than the round sticks, tied at the joints by cords.

On paper, Willard had performed the painstaking task of dimensioning every stone, each with its top (the build) a little narrower than its bottom (the bed), so that the true taper of the obelisk would be maintained. To set the stone to line was the chore of the erecting force. On the top of the stone already in place lime mortar was spread, enriched with hydraulic cement, and with a sprinkling of iron filings. (A popular, modern, commercial waterproofing compound utilizes the fact that, upon oxidizing, iron particles mixed with the cement mortar expand, thus reducing the voids and producing a denser mass when the mortar sets.)

Temporary wooden wedges would be placed at the corners of the stone already in place, to support the heavy new stone until the mortar had hardened. When the ponderous stone had been speedily raised from the ground to a level a few inches above the mortar, the engine would be stopped; the derrick adjusted to right or left, and in or out, until the stone was very closely in line. Next, riggers would push on the bridle chain which attached to the two lewises in the stone's top surface, guiding it to its true position in its gentle descent as the engine lowered it the few inches needed to bed it. A tiny fraction of an inch "out-of-line" would be serious, for such errors if repeated, or if not compensating, would visibly throw the monument out of line. On a light stage high above the ground, bracing themselves against gusts of wind, the riggers would be intent on the necessity for such accuracy, but not forgetting their own safety, for no careless workman could work for Savage. They would remember that, while laying the last stone of the 12th course, at the southwest corner, one man had been pushed off (the only death on the project).

Profitable Project

Happily, the good riggers raised their monument, course by course, to the top. They were under a boss who knew his trade, and he was making money—the sign of content on any construction project. (Savage made some more money after the monument was finished, when he retained the steam engine that had been used for hoisting stone, for the purpose of raising a passenger car to the top. For the car ride he collected $0.20, as against the $0.125 for visitors who climbed the stairs.)

The practical riggers would not be disturbed at the proposal, advanced when the monument neared its top, that the apex be modified to form a platform to accommodate visitors. Aesthetic Bostonians were much disturbed at this proposal. Happily, the Bunker Hill Monument Association voted down this architectural atrocity.

High Capstone

On Saturday, 23 July 1842, several hundred of our early rising Bostonian ancestors rose earlier than usual to arrive at the monument at 6:00 A.M. On the ground at the base, they studied the capstone; a small stone pyramid, three feet, six inches high, stoutly lashed to the derrick hook, and with an American flag at the top. Standing on the capstone, firmly grasping the hoisting rope, Colonel Charles R. Carnes waited for the signal to hoist. When the clock struck six, a signal gun was fired, and the capstone, bearing the good colonel, started up. In 16 minutes it had reached the top; at 6:30 A.M. it had been bedded, and a national salute announced to all Boston that the Bunker Hill Monument was completed.

By railroad, great multitudes came to Boston for the dedication of the Bunker Hill Monument, nearly a year later, on 17 June 1843. Unlike the time of 18 years earlier, when the cornerstone had been laid, stagecoaches were not the main conveyance for visitors to Boston. Indeed they were decidedly on the way out, and would soon be but symbols of an era of traveling discomfort, as the railroad completely took over. President Tyler and his cabinet attended; Daniel Webster matched with sonorous eloquence his famous speech at the laying of the cornerstone, and there were still 13 very aged surviving veterans of the battle able to attend.

Foucault's Experiment

Seven years later (1850), two Harvard professors checked with elaborate apparatus, paid for by members of the Bunker Hill Monument Association and the Massachusetts Charitable Association, the famous experiment of Foucault which used a long pendulum to prove the daily rotation of the earth. Suspended in the newel by an annealed wire, 210 feet long, the oscillations of a pointer, attached to a 31-pound British cannon ball relic of the battle, were observed; and its plane of swing was seen to revolve during the day from right to left of the observer. A sudden shower on a previously bright day complicated the experiment, until Professor Eben N. Horsford discovered the reason. Cooled by the rain the monument's exposed face contracted; its apex moved correspondingly and carried the point of suspension of the pendulum with it. As observed years later on the Washington Monument, Horsford deduced that the side of an obelisk, exposed to the hot sun, expanded, and that its apex followed the sun in the sun's travel from east to west. Such motion is tiny, and the ingenuity of the apparatus to observe it was notable. The path of the orbit of the bob, registering both the earth's rotation and the effect of varying heat on the monument's sides, was described as an irregular ellipse with major axis of one-half an inch.

Today

Two hundred thousand persons visit Bunker Hill every year. Of these visitors 20,000 pay their $0.10, and presumably climb the 294 granite steps to the top.[12] Very few Bostonians are among these visitors.[13] They are of various types: honeymooners and casual tourists, whose list of the sights to be seen in historic Boston includes the monument, and historically minded youngsters, one of whom was recently caught in a heated argument with his father as to where the order was to "wait 'til you can see the whites of their eyes." Surely, the stout young man who recently lugged 25 pounds to the top—his young daughter—was not typical.

[12] Estimate of custodian from a review of his register, 1951.

[13] “Those pangs of conscience he feels every June 17 are as close as the average Bostonian ever comes to climbing Bunker Hill Monument.”—*Boston Globe*, 18 June 1951.

The evidence that the monument is probably the most popular of the historic shrines of old Boston would be as pleasing to the charter members of the Bunker Hill Monument Association as to the present members of this venerable society, which lists among present and past members, 30 generals, 12 admirals, 12 Presidents of the United States, a score of Massachusetts governors, 20 mayors of Boston, and 6 presidents of Harvard University. It is a healthful society to belong to, the 1949 *Proceedings* say, in the mention of 46 half-century members of whom 16 were then living.

Many years ago, the Association voted to hold patriotic exercises every year at the monument, and this resolution has been faithfully fulfilled. The annual ceremonies today are very different from those of the earlier years; they have followed the varying pattern in which American citizens have celebrated their national anniversaries during the more than a century since the monument was dedicated in 1843. They were solemn occasions during the earlier years. In what better spot could the Yankees of the trying days of the Civil War compare their convictions with those who fought for similar principles than at the monument, as they listened to the stirring eloquence of their War Governor, John A. Andrew. In later years the holiday spirit took over, with a fireman’s muster to please the older folk, while the youngsters of Greater Boston made Bunker Hill Day on 17 June a parallel in firecracker noise and casualties to fingers and eyesight, to the Fourth of July, of which it was a preview. Today, Bunker Hill Day is a huge neighborhood festival, with block parties and a skillfully routed parade which seems to pass every house on the hill. They who enjoy things most are the children of working people, not of the wealthy families that once lived in the sightly dwellings of Bunker Hill. Each boy or girl can give a visitor the story of the battle in detail, and recite the precise dimensions of the monument. (They collect from tourists for this information, for it is one way by which Charlestown youngsters get their spending money.) These children would rarely answer to the old names: Prescott, Warren, Putnam, or similar Yankee names. They are mostly of second or third generation European families: proud Americans, fortunate to live near the site of one of America's most famous historical shrines. Their festival is a heartening occasion to witness, for it is American democracy at its best.

Through it all the monument rises above its unadorned settings; except that the crown of the hill has been removed, it could still be the New England hilltop farm on which the battle was fought. The obelisk rises in the simplicity of its straight lines and clean angles, with no curves, and with the somber gray of its harsh-textured masonry unrelieved by any greenery of foundation shrubbery. The rugged monument is symbolic of the stern spirit of those who fought in the battle, and of the determination of those who solved the problem of building this massive memorial to them, in the pioneer days of American architecture and engineering.

Transcriber’s Notes

Punctuation, hyphenation, and spelling were made consistent when a predominant preference was found in the original book; otherwise they were not changed.

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